# WASP-43b

R-1901

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_190309` · created 2026-08-03T14:07:58+00:00

**BENCHMARK: ACCURATE — fit consistent with published values**

## Results

| | |
|---|---|
| **Mid-Transit Time (Tmid)** | 2458551.7372 +/- 0.0014 BJD_TDB |
| **Ratio of Planet to Stellar Radius (Rp/R*)** | 0.198 +/- 0.025 |
| **Transit depth (Rp/Rs)^2** | 3.91 +/- 0.97 [%] |
| **Orbital Inclination (inc)** | 78.66 +/- 0.71 |
| **Airmass coefficient 1 (a1)** | 0.984 +/- 0.011 |
| **Airmass coefficient 2 (a2)** | 0.012 +/- 0.015 |
| **Scatter in the residuals of the lightcurve fit is** | 1.14 % |
| **Best Comparison Star** | #3 - [379, 142] |
| **Optimal Aperture** | 3.6 |
| **Optimal Annulus** | 14.81 |
| **Transit Duration (day)** | 0.0393 +/- 0.0049 |

## Accuracy vs published ephemeris (ExoClock/NEA)

| | |
|---|---|
| **Rp/R■ fit vs published** | 0.198 ± 0.025 vs 0.1594 (deviation 1.16σ) |
| **Duration fit vs published** | 0.0393 d vs 0.0512 d (deviation 1.83σ) |
| **Mid-time O–C** | -1.6 min |
| **Depth SNR** | 6.0 |
| **Residual scatter** | 1.14 % |

## Light curve

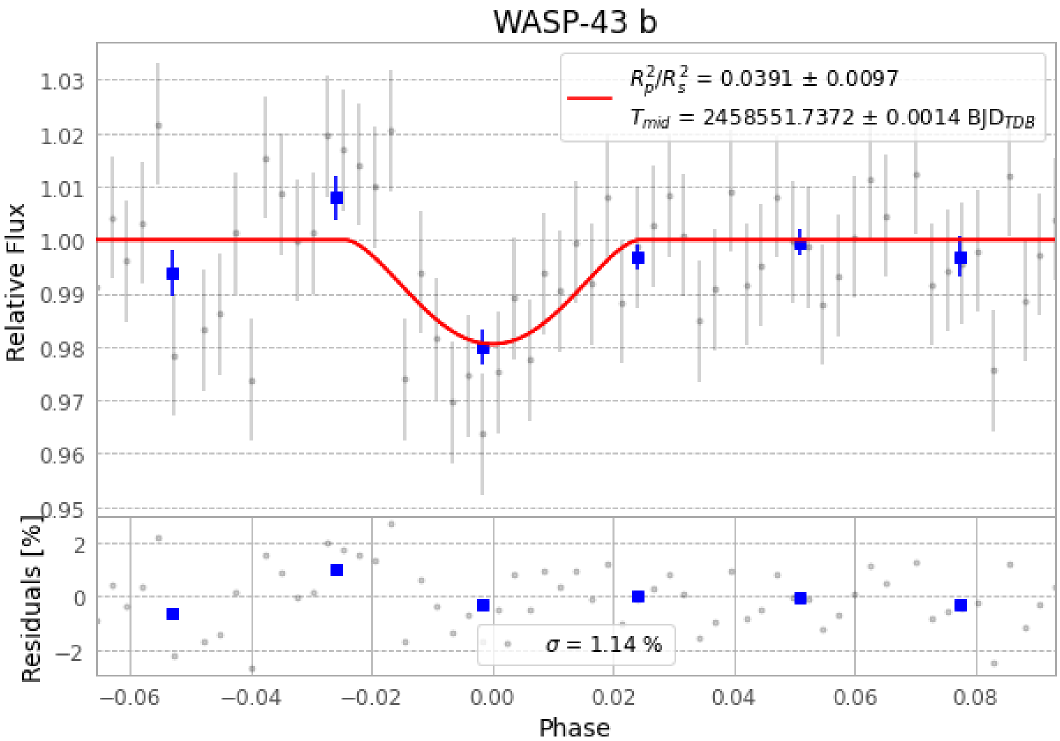


Figure 1 — FinalLightCurve_WASP-43 b_2019-03-08.png (SHA-256 c9986f5ff448d959…)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [446, 144], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 1093813a6c0e61d7…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

## Data provenance

63 FITS frames (41 MB), `WASP-431903090415 12.FITS` → `WASP-43190309072112.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-190309.log (addec2528f46…), FinalLightCurve_WASP-43 b_2019-03-08.png (c9986f5ff448…), FinalLightCurve_WASP-43 b_2019-03-08.csv (9dece11df734…)

## Compute resources

Wall time 200.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

---

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 14:07Z.